

UNIVERSITÄT
LEIPZIG

International Physics Studies Program

Advanced Physics Laboratory

Optical Spectroscopy of Colour Centres and Molecules

Franck–Condon Principle, Iodine Vapour, and F-centres

Group: O4

Date: 09 June 2026

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Abstract

This experiment investigated optical spectroscopy of molecular iodine vapour and colour centres in alkali halide crystals using a Lambda 40 UV/VIS spectrometer. The spectrometer was first calibrated with a holmium oxide filter. A scan speed of 480 nm/min and a slit width of 0.5 nm were selected, giving an average relative wavelength error of approximately 0.39%. Transmission measurements of glass and quartz cuvettes, water, colour filters, and an interference filter showed the expected wavelength-dependent behaviour, including improved ultraviolet transmission of quartz, a water absorption band near 975 nm, colour-filter maxima at 466 nm, 546 nm, and 644 nm, and a blue shift of the interference-filter maximum with increasing angle of incidence. The iodine vapour spectrum showed vibronic absorption structure in the visible region. Using a reference-assisted line assignment and an anharmonic oscillator fit, the electronic transition energy was determined as $E_{\text{electr}} = (1.9389 \pm 0.0003)$ eV. The convergence energy was $E_K = 2.4947$ eV, giving a ground-state dissociation energy of $E_D'' = 1.5547$ eV and an excited-state dissociation energy of $E_D' = 0.5558$ eV. F-centre absorption bands were observed in UV-irradiated KBr and X-ray irradiated NaCl. For KBr, the main absorption maximum was initially found near 640 nm, corresponding to an excitation energy of approximately 1.937 eV. The F-centre concentration decreased from $2.51 \times 10^{14}\ 1/\text{cm}^3$ directly after irradiation to $7.74 \times 10^{13}\ 1/\text{cm}^3$ after 45 min. For NaCl, the absorption maximum was found at 473.65 nm, corresponding to $E_m = 2.6176$ eV, with a linewidth of $\Delta E_H = 0.4629$ eV and a concentration of $1.01 \times 10^{15}\ 1/\text{cm}^3$. The comparison with classical and quantum F-centre models showed that both models reproduced the correct qualitative trend of higher excitation energy for smaller lattice constants, while the classical oscillator model gave better quantitative agreement than the simple quantum box model.

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1. Introduction

Optical spectroscopy is a powerful method for studying electronic transitions in atoms, molecules, and solids. In the UV/VIS range, the interaction of light with matter can reveal absorption bands, transmission windows, defect states, and the coupling between electronic and vibrational degrees of freedom. This experiment investigated two apparently different systems: molecular iodine vapour and colour centres in alkali halide crystals. Although iodine vapour belongs to molecular spectroscopy and colour centres belong to solid-state physics, both systems are governed by electronic transitions that are strongly influenced by the motion of atoms or ions. This common physical basis is described by the Franck–Condon principle, according to which an electronic transition occurs much faster than the nuclei or surrounding lattice ions can rearrange their positions [1].

The molecular part of the experiment focused on the transmission spectrum of iodine vapour, I_2. In a diatomic molecule, electronic excitation is accompanied by changes in the vibrational state of the molecule. Since the electronic and vibrational energy scales differ strongly, the observed spectrum contains a structured sequence of absorption lines rather than a single isolated electronic transition. The positions of these absorption minima can be used to analyse the vibrational progression of the excited electronic state. From the convergence behaviour of the absorption lines, quantities such as the convergence energy, the dissociation energies of the ground and excited states, and the electronic transition energy can be determined [1].

The solid-state part of the experiment dealt with colour centres, especially F-centres, in alkali halide crystals such as KBr and NaCl. An F-centre consists of an anion vacancy that traps an electron. This trapped electron forms a localized defect state inside the wide band gap of the otherwise transparent ionic crystal. Optical excitation of this electron produces a broad absorption band in the visible spectral range, which gives the crystal its characteristic colour after irradiation. In the experiment, F-centres were generated in KBr by UV irradiation, while pre-irradiated NaCl crystals were investigated for comparison. The measured transmission spectra were used to determine the excitation energy, linewidth, and concentration of the colour centres [1].

Before analysing the molecular and defect spectra, the performance of the UV/VIS spectrometer was checked using a holmium oxide filter with known absorption bands. The influence of instrumental parameters such as slit width and scanning speed was examined because these settings affect spectral resolution, peak shape, and the reliability of extracted wavelength positions. Transmission spectra of glass and quartz cuvettes, water-filled cuvettes, colour filters, and an interference filter were also recorded in order to study how optical materials modify the transmitted spectrum. These measurements provided a calibration and reference basis for the later analysis of iodine vapour and colour-centre absorption bands.

All spectra were recorded using a Lambda 40 double-beam UV/VIS spectrometer. The experiment therefore combined instrumental spectroscopy, molecular absorption spectroscopy, and defect-state spectroscopy in solids. The main goal was to connect the measured transmission spectra with the underlying microscopic physics: vibrationally resolved electronic transitions in iodine, localized electronic defect states in alkali halides, and the Franck–Condon-type coupling between electronic excitation and atomic or lattice motion.

2. Theoretical Background

2.1 Transmission, absorbance and absorption coefficient

Optical transmission spectroscopy measures how strongly a sample attenuates light as a function of wavelength. The transmission is defined as the ratio of transmitted intensity to incident intensity,

$$T(\lambda) = \frac{I_t(\lambda)}{I_0(\lambda)}, \tag{1}$$

where $I_0(\lambda)$ is the incident intensity and $I_t(\lambda)$ is the transmitted intensity. In the Lambda 40 double-beam spectrometer, this ratio was obtained by comparing the sample beam with a reference beam. A reference measurement without the absorbing sample, or with the corresponding empty sample holder, was used to define the baseline transmission.

The absorbance is commonly defined as

$$A(\lambda) = -\log_{10} T(\lambda). \tag{2}$$

Absorbance is especially useful when absorption bands are discussed, because transmission minima correspond to absorbance maxima. For a homogeneous absorbing medium of thickness d, neglecting reflection losses at the surfaces, the Beer–Lambert law gives

$$I_t = I_0 \exp(-\alpha d), \tag{3}$$

or equivalently,

$$T = \exp(-\alpha d), \tag{4}$$

where α is the absorption coefficient. Solving Eq. (4) for α gives

$$\alpha = -\frac{1}{d} \ln T. \tag{5}$$

If a reference transmission T_0 was used to remove the slowly varying background, the absorption coefficient of the additional absorption band was calculated from

$$\alpha(\lambda) = \frac{1}{d} \ln\left(\frac{T_0(\lambda)}{T(\lambda)}\right). \tag{6}$$

For plane-parallel crystals, reflection at the front and back surfaces can also influence the measured transmission. In this case, the transmission is more accurately described by

$$T = \frac{(1-R)^2 \exp(-\alpha d)}{1 - R^2 \exp(-2\alpha d)}, \tag{7}$$

where R is the reflectivity of one crystal surface. For normal incidence, the reflectivity is

$$R = \left(\frac{n-1}{n+1}\right)^2, \tag{8}$$

with n the refractive index of the crystal. For NaCl and KBr, the reflectivity is relatively small, so the Beer–Lambert approximation in Eq. (4) is often sufficient for a first analysis.

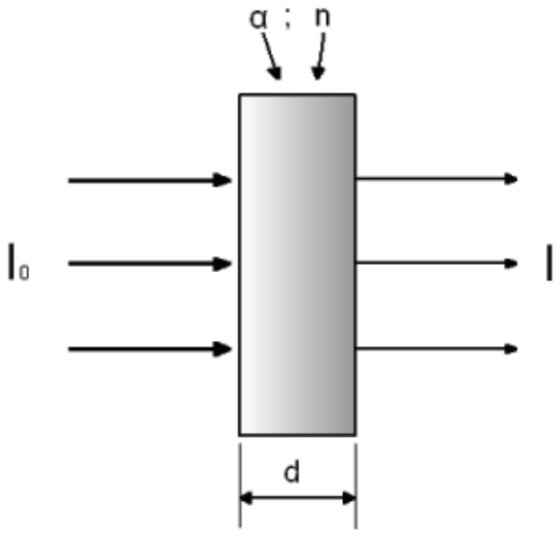


Figure 1: Transmission through a plane-parallel crystal of thickness d with absorption coefficient α and refractive index n. The measured transmission depends on absorption inside the crystal and, more generally, on reflection losses at the two crystal surfaces.

2.2 Photon energy, wavelength and wavenumber

The photon energy is related to the wavelength by

$$E = \frac{hc}{\lambda}, \tag{9}$$

where h is Planck's constant and c is the speed of light. In spectroscopy it is also common to use the wavenumber,

$$\tilde{\nu} = \frac{1}{\lambda}. \tag{10}$$

The photon energy can then be written as

$$E = hc\tilde{\nu}. \tag{11}$$

When the wavelength is given in nanometres and the energy in electronvolts, Eq. (9) becomes

$$E, /, \mathrm{eV} = \frac{1239.841984}{\lambda, /, \mathrm{nm}}. \tag{12}$$

If the wavenumber is given in $1/\mathrm{cm}$, the corresponding energy in electronvolts is

$$E, /, \mathrm{eV} = \frac{\tilde{\nu}, /, 1/\mathrm{cm}}{8065.54}. \tag{13}$$

These conversions were used to transform measured absorption wavelengths into transition energies.

2.3 Optical filters and instrumental broadening

The measured spectrum is influenced not only by the sample but also by the instrumental settings of the spectrometer. The slit width controls the spectral bandwidth of the light reaching the sample and detector. A smaller slit width improves spectral resolution and makes narrow absorption features sharper, but it also reduces the transmitted intensity. A larger slit width increases the intensity but broadens spectral features and can make closely spaced absorption lines more difficult to resolve.

The scanning speed also affects the quality of a recorded spectrum. At very high scanning speeds, narrow absorption minima can be distorted or broadened because the instrument samples the spectral feature over a shorter time. For this reason, the holmium oxide filter was used as a wavelength calibration standard and as a practical test of the chosen slit width and scanning speed.

Colour filters transmit selected wavelength regions and absorb or suppress others. Therefore, the perceived colour of a filter is connected to the spectral range where its transmission is largest. An interference filter works differently: it transmits wavelengths that satisfy a constructive-interference condition inside a thin-film multilayer structure. For normal incidence, the approximate condition for constructive transmission can be written as

$$2n_{\mathrm{eff}} d_{\mathrm{eff}} = m\lambda_0, \tag{14}$$

where n_{eff} is an effective refractive index, d_{eff} is the effective optical layer thickness, m is the interference order, and λ_0 is the central transmission wavelength at normal incidence. At oblique incidence, the optical path length inside the filter changes. A commonly used approximate expression for the angular shift of the transmission maximum is

$$\lambda(\theta) = \lambda_0 \sqrt{1 - \frac{\sin^2\theta}{n_{\mathrm{eff}}^2}}, \tag{15}$$

where θ is the external angle of incidence. Equation (15) shows that the transmission maximum shifts to shorter wavelengths as the angle of incidence increases. This corresponds to the blue shift observed for the interference filter.

2.4 Vibrational motion of a diatomic molecule

The iodine molecule, I_2, is a diatomic molecule. Its nuclear motion can first be approximated by a harmonic oscillator. If the two atoms have masses m_1 and m_2, the reduced mass is

$$\mu = \frac{m_1 m_2}{m_1 + m_2}. \tag{16}$$

For small displacements from the equilibrium bond length, the restoring force is approximately proportional to displacement. The vibrational angular frequency is

$$\omega_{\mathrm{s}} = \sqrt{\frac{k}{\mu}}, \tag{17}$$

where k is the force constant of the molecular bond. The corresponding vibrational wavenumber is

$$\tilde{\nu}_{\mathrm{s}} = \frac{1}{2\pi c}\sqrt{\frac{k}{\mu}}. \tag{18}$$

In the quantum-mechanical harmonic oscillator, the allowed vibrational energies are

$$E_n = hc\tilde{\nu}_{\mathrm{s}}\left(n + \frac{1}{2}\right), \qquad n = 0, 1, 2, \ldots. \tag{19}$$

The lowest vibrational state has the zero-point energy

$$E_0 = \frac{1}{2}hc\tilde{\nu}_{\mathrm{s}}. \tag{20}$$

The harmonic oscillator approximation predicts equally spaced vibrational energy levels. Real molecules, however, are anharmonic. The spacing between neighbouring vibrational levels decreases as the molecule approaches dissociation. A more realistic potential is the Morse potential,

$$U(x) = E_{\mathrm{D}}\left(1 - \exp(-ax)\right)^2, \tag{21}$$

where E_{D} is the dissociation energy and a describes the steepness of the potential. The corresponding vibrational energies, keeping the leading anharmonic correction, are

$$E_n = hc\tilde{\nu}_{\mathrm{s}}\left[\left(n + \frac{1}{2}\right) - y_{\mathrm{e}}\left(n + \frac{1}{2}\right)^2\right], \tag{22}$$

where y_{e} is the anharmonicity constant. The negative quadratic term causes the level spacing to decrease for larger n, which is essential for determining the convergence limit of the iodine spectrum.

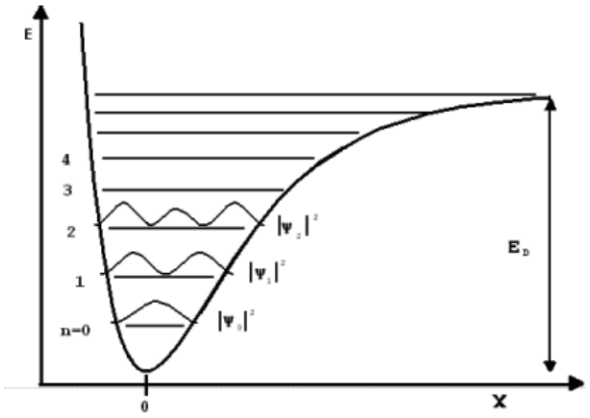


Figure 2: Anharmonic molecular potential with vibrational energy levels and probability density $|\Psi_n|^2$. In contrast to the harmonic oscillator, the spacing between neighbouring levels decreases as the dissociation limit is approached.

2.5 Franck–Condon principle

Electronic transitions occur on a much shorter time scale than nuclear motion. During an electronic excitation, the positions and momenta of the nuclei therefore remain approximately unchanged. In a potential-energy diagram, this means that electronic transitions are represented by nearly vertical transitions. This is the Franck–Condon principle.

The transition probability between an initial vibrational state Ψ'' and a final vibrational state Ψ' is governed by the overlap of the two vibrational wave functions. The relevant overlap integral is

$$P \propto \left| \int \Psi'^{*}\Psi'' \, \mathrm{d}r \right|^2. \tag{23}$$

A transition is therefore most intense when the vibrational wave functions in the lower and upper electronic states overlap strongly. For iodine vapour this produces a sequence of vibrationally resolved absorption lines. For F-centres, the same principle explains why a broad absorption band is observed instead of an infinitely sharp electronic line: the electronic excitation is coupled to the motion of the surrounding lattice ions.

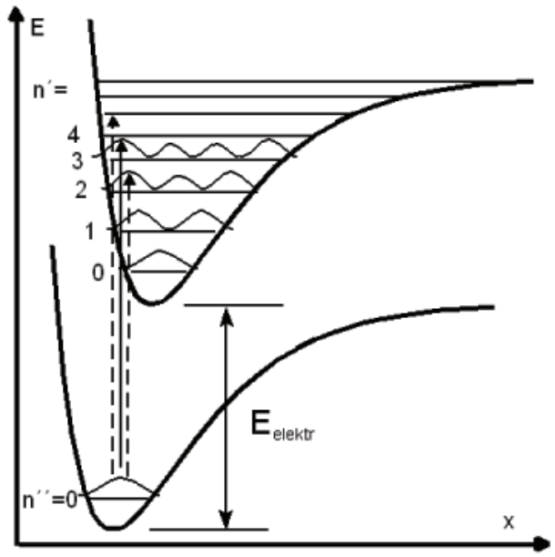


Figure 3: Franck–Condon picture of electronic-vibrational transitions. Since electronic excitation is much faster than nuclear motion, the transition occurs approximately vertically in the potential-energy diagram.

2.6 Iodine vapour spectrum and dissociation energies

The visible absorption spectrum of iodine vapour consists of electronic-vibrational transitions. The lower electronic state is described by vibrational quantum numbers n'', while the upper electronic state is described by n'. In the experiment, the observed absorption minima were treated as transitions starting mainly from the vibrational ground state of the electronic ground state, $n'' = 0$, to different vibrational levels n' of the excited electronic state.

For transitions from $n'' = 0$, the transition energy can be written approximately as

$$\Delta E(n') = E_{\mathrm{electr}} + E'_{\mathrm{s}}\left(n' + \frac{1}{2}\right) - E'_{\mathrm{s}}y'_{\mathrm{e}}\left(n' + \frac{1}{2}\right)^2, \quad (24)$$

where E_{electr} is the electronic transition energy between the minima of the two electronic potential curves, $E'_{\mathrm{s}} = hc\tilde{\nu}'_{\mathrm{s}}$ is the vibrational energy spacing of the excited electronic state, and y'_{e} is the anharmonicity constant of the excited state.

The energy spacing between neighbouring absorption lines is

$$D_{\mathrm{L}}(n') = \Delta E(n'+1) - \Delta E(n'). \quad (25)$$

Using Eq. (24), this becomes

$$D_{\mathrm{L}}(n') = E'_{\mathrm{s}}\left[1 - 2y'_{\mathrm{e}}(n'+1)\right]. \quad (26)$$

Thus, the line spacing decreases linearly with increasing n'. At the convergence limit, the spacing between neighbouring lines tends to zero. Setting $D_{\mathrm{L}} = 0$ gives the effective convergence quantum number,

$$n'_{\mathrm{E}} = \frac{1}{2y'_{\mathrm{e}}} - 1 \quad (27)$$

The convergence energy is then obtained by substituting n'_{E} into the transition-energy expression,

$$E_{\mathrm{K}} = E_{\mathrm{electr}} + E'_{\mathrm{s}}\left(n'_{\mathrm{E}} + \frac{1}{2}\right) - E'_{\mathrm{s}}y'_{\mathrm{e}}\left(n'_{\mathrm{E}} + \frac{1}{2}\right)^2. \quad (28)$$

If E_{K}, E'_{s}, and y'_{e} are known, the electronic transition energy can also be written as

$$E_{\mathrm{electr}} = E_{\mathrm{K}} - E'_{\mathrm{s}}\left(n'_{\mathrm{E}} + \frac{1}{2}\right) + E'_{\mathrm{s}}y'_{\mathrm{e}}\left(n'_{\mathrm{E}} + \frac{1}{2}\right)^2. \quad (29)$$

The dissociation energies are obtained from the energy-level scheme of the iodine molecule. The ground-state dissociation energy is

$$E''_{\mathrm{D}} = E_{\mathrm{K}} - E_{\mathrm{A}}, \quad (30)$$

where E_{A} is the atomic excitation energy of iodine. The excited-state dissociation energy is

$$E'_{\mathrm{D}} = E_{\mathrm{K}} - E_{\mathrm{electr}}. \quad (31)$$

These relations were used to determine the convergence energy, the electronic transition energy, and the dissociation energies from the iodine absorption spectrum.

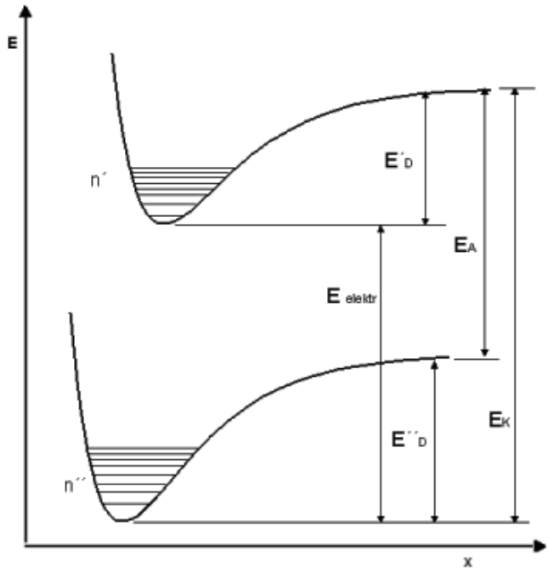


Figure 4: Energy scheme used for the iodine dissociation analysis. The convergence energy, electronic transition energy, and dissociation energies are related through the potential curves of the ground and excited electronic states.

2.7 Colour centres in alkali halide crystals

Alkali halide crystals such as NaCl and KBr have a large band gap and are therefore normally transparent in the

visible spectral range. Irradiation with UV light, X-rays, electrons, or gamma radiation can create point defects in the crystal lattice. One important defect type is the F-centre. An F-centre is an anion vacancy occupied by an electron. In NaCl, this corresponds to a missing Cl^- ion with a trapped electron at the vacancy; in KBr, it corresponds to a missing Br^- ion with a trapped electron.

The trapped electron forms localized energy states inside the forbidden band gap. Optical excitation from the ground state to an excited state of the trapped electron produces an absorption band in the visible range. Because visible wavelengths are removed from the transmitted light, the irradiated crystal appears coloured.

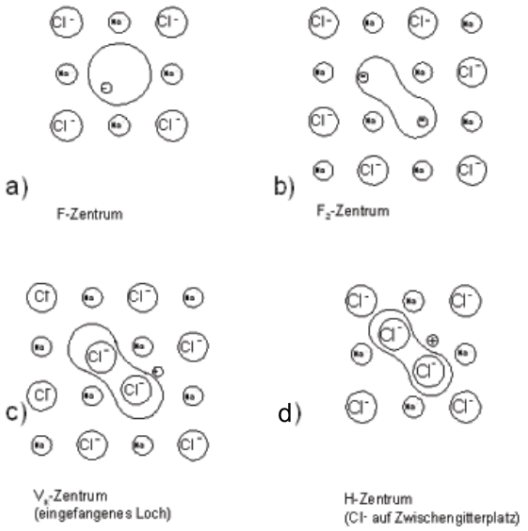


Figure 5: Defect structures in alkali halides: (a) halogen ion vacancy with a trapped electron, corresponding to an F-centre; (b) two neighbouring colour centres; (c) self-trapped hole; (d) halogen ion at an interstitial site, forming a Frenkel pair.

The maximum of the F-centre absorption band defines the excitation energy,

$$E_{\mathrm{m}} = \frac{hc}{\lambda_{\mathrm{m}}}, \tag{32}$$

where λ_{m} is the wavelength at the absorption maximum. The width of the absorption band was described by the full width at half maximum. If the two half-maximum wavelengths are λ_1 and λ_2, the corresponding energy linewidth is

$$\Delta E_{\mathrm{H}} = \left| \frac{hc}{\lambda_1} - \frac{hc}{\lambda_2} \right|. \tag{33}$$

The broad linewidth of the F-centre absorption band is a consequence of electron–phonon coupling. The electronic transition changes the equilibrium positions of the surrounding ions, so many vibrational configurations contribute to the absorption process.

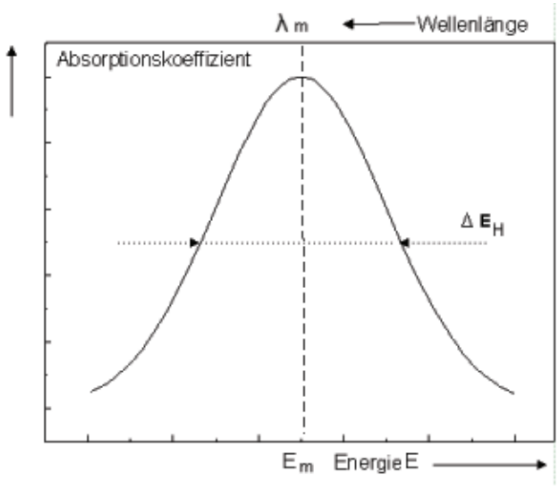


Figure 6: Schematic absorption band of an F-centre. The maximum defines the excitation energy E_{m}, while the full width at half maximum gives the linewidth ΔE_{H}.

2.8 Franck–Condon interpretation of the F-centre linewidth

Although an F-centre has discrete electronic states, its optical absorption does not appear as a sharp line. The trapped electron interacts strongly with the surrounding lattice ions. The total energy of the defect therefore depends on a configurational coordinate R, which represents the displacement of the neighbouring ions from the defect centre.

In the ground state, the surrounding ions have an equilibrium configuration R_0. After electronic excitation, the electron distribution changes and the equilibrium configuration shifts to a different value R_{f}. Since the electronic transition is much faster than the motion of the ions, the transition occurs vertically in the configurational-coordinate diagram. The absorption energy is therefore distributed over a range of vibrational final states, producing a broad absorption band. This is the same Franck–Condon principle that explains the vibrational progression in iodine, but applied to a localized defect in a crystal lattice.

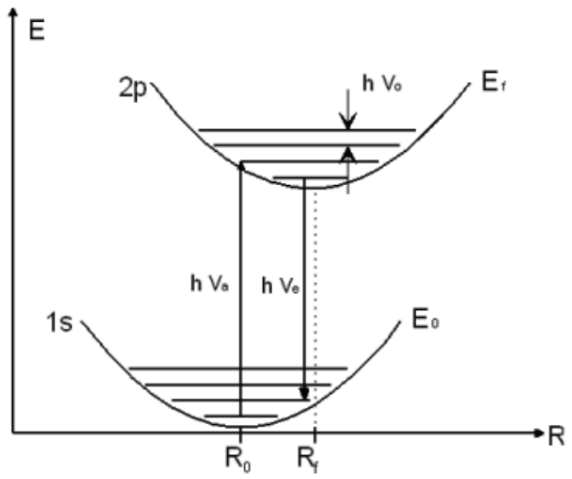


Figure 7: Configurational-coordinate picture of F-centre absorption and emission. The displacement between the ground- and excited-state equilibrium positions causes broad optical bands.

2.9 Classical model of the F-centre

In the classical model, the electron trapped in the anion vacancy is treated as an oscillator. The missing negative ion leaves an effective positive charge distribution formed by the surrounding cations. This positive charge is approximated as being homogeneously distributed over a sphere of radius R. The radius is related to the lattice constant a by

$$R^3 = \frac{3\sqrt{2}}{16\pi} a^3. \tag{34}$$

For an electron displaced by a distance r from the centre of this charge distribution, the restoring force is proportional to r,

$$F_r = -\frac{e^2}{4\pi\varepsilon\varepsilon_0}\frac{r}{R^3}, \tag{35}$$

where e is the elementary charge, ε_0 is the vacuum permittivity, and ε is the relative permittivity used in the model. The equation of motion is therefore

$$m_\mathrm{e}\ddot{r} = -\frac{e^2}{4\pi\varepsilon\varepsilon_0}\frac{r}{R^3}. \tag{36}$$

Comparing Eq. (36) with the harmonic oscillator equation gives

$$\omega^2 = \frac{e^2}{4\pi\varepsilon\varepsilon_0 m_\mathrm{e} R^3}. \tag{37}$$

The classical excitation energy is then

$$E_\mathrm{cl} = \hbar\omega = \hbar\sqrt{\frac{e^2}{4\pi\varepsilon\varepsilon_0 m_\mathrm{e} R^3}}. \tag{38}$$

This model gives a simple physical picture of the trapped electron as a harmonically bound charge. However, it is only an approximation because it replaces the discrete ionic lattice and the real defect potential by a continuous spherical charge distribution.

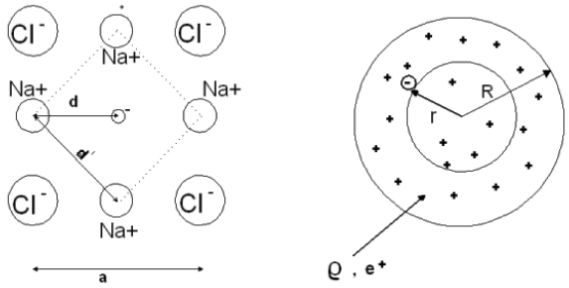


Figure 8: Classical model of an F-centre. The trapped electron is treated as a harmonically bound charge inside an effective positive charge distribution produced by the surrounding ions.

2.10 Quantum-mechanical box model of the F-centre

A more quantum-mechanical description treats the trapped electron as a particle confined inside a three-dimensional potential box. In the simplest model, the potential is zero inside a rectangular box and infinite outside,

$$V(x,y,z) = \begin{cases} 0, & 0 \le x \le a,\ 0 \le y \le b,\ 0 \le z \le c, \\ \infty, & \text{otherwise.} \end{cases} \tag{39}$$

Inside the box, the time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m_\mathrm{e}}\nabla^2\Psi = E\Psi. \tag{40}$$

Solving this equation with the boundary condition that the wave function vanishes at the box walls gives the energy eigenvalues

$$E_{n_1,n_2,n_3} = \frac{\hbar^2\pi^2}{2m_\mathrm{e}}\left(\frac{n_1^2}{a^2} + \frac{n_2^2}{b^2} + \frac{n_3^2}{c^2}\right), \tag{41}$$

where n_1, n_2, and n_3 are positive integers. For a cubic box with $a = b = c$, this simplifies to

$$E_{n_1,n_2,n_3} = E_0\left(n_1^2 + n_2^2 + n_3^2\right), \tag{42}$$

with

$$E_0 = \frac{h^2}{8m_\mathrm{e}a^2}. \tag{43}$$

The ground state corresponds to $(n_1, n_2, n_3) = (1, 1, 1)$ and has energy

$$E_{111} = 3E_0. \tag{44}$$

The first excited states correspond to permutations of $(2, 1, 1)$ and have energy

$$E_{211} = E_{121} = E_{112} = 6E_0. \tag{45}$$

The transition energy from the ground state to the first excited state is therefore

$$E_{\mathrm{qm}} = 6E_0 - 3E_0 = 3E_0 = \frac{3h^2}{8m_{\mathrm{e}}a^2}. \tag{46}$$

This expression predicts that the excitation energy decreases with increasing lattice constant. The model is consistent with the empirical trend that larger alkali halide lattice constants produce lower F-centre excitation energies.

Equation (46) can also be inverted to estimate an effective lattice constant from the measured excitation energy,

$$a_{\mathrm{exp,qm}} = \sqrt{\frac{3h^2}{8m_{\mathrm{e}}E_{\mathrm{m}}}}. \tag{47}$$

The quantum box model is still strongly simplified, because the real F-centre potential is neither infinitely deep nor perfectly cubic. It also neglects lattice polarization and detailed electron–phonon coupling. Nevertheless, it provides a useful comparison with the classical oscillator model.

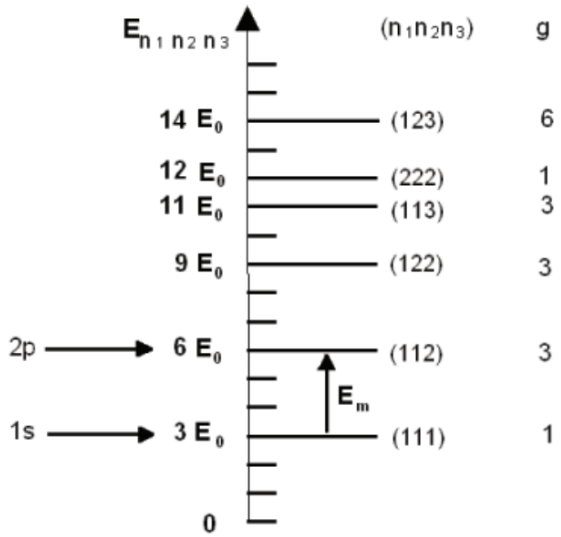


Figure 9: Energy states of the cubic box model. The optical excitation of the trapped electron is approximated as a transition from the 1s-like ground state to the first 2p-like excited state.

2.11 Concentration of F-centres

The concentration of F-centres is related to the integrated absorption strength of the F-centre band. In the Lorentz-oscillator approximation, the integral absorption is

$$\int \alpha(\nu), \mathrm{d}\nu = \frac{e^2 N}{4c^2\varepsilon_0 n m_{\mathrm{e}}} \left(\frac{n^2+2}{3}\right)^2, \tag{48}$$

where N is the F-centre concentration, n is the refractive index, and ν is the optical frequency. Solving for N gives

$$N = \frac{4c^2\varepsilon_0 n m_{\mathrm{e}}}{e^2} \left(\frac{3}{n^2+2}\right)^2 \int \alpha(\nu), \mathrm{d}\nu. \tag{49}$$

If the absorption spectrum is integrated over wavenumber instead of frequency, the relation $\nu = c\tilde{\nu}$ must be used. Since

$$\mathrm{d}\nu = c, \mathrm{d}\tilde{\nu}, \tag{50}$$

the concentration can also be written as

$$N = \frac{4c^3\varepsilon_0 n m_{\mathrm{e}}}{e^2} \left(\frac{3}{n^2+2}\right)^2 \int \alpha(\tilde{\nu}), \mathrm{d}\tilde{\nu}, \tag{51}$$

provided that SI units are used consistently. If $\tilde{\nu}$ is integrated in $1/\mathrm{cm}$ and α is given in $1/\mathrm{cm}$, the unit conversion has to be treated consistently before reporting N in $1/\mathrm{cm}^3$.

The concentration analysis therefore required three steps. First, the absorption coefficient was calculated from the transmission spectrum using Eq. (6) or Eq. (5). Second, the F-centre absorption band was integrated over the selected spectral range. Third, the integral absorption was converted into a defect concentration using Eq. (49) or Eq. (51).

3. Experimental Setup and Methods

3.1 UV/VIS spectrometer

All transmission spectra were recorded using a Lambda 40 double-beam UV/VIS spectrometer. The instrument covered a wavelength range from approximately 190 nm to 1100 nm, which allowed measurements from the near-ultraviolet through the visible region and into the near-infrared. The light source system consisted of a deuterium lamp for the ultraviolet range and a second lamp for the visible and near-infrared range. The spectral bandwidth could be adjusted by changing the slit width, and the scanning speed could be selected through the spectrometer software.

The optical system of the spectrometer contained a filter wheel, entrance slit, monochromator, exit slit, beam splitter, sample path, reference path, and two photodetectors. The monochromator selected the wavelength incident on the sample by means of a dispersive grating. The beam splitter separated the monochromatic light into a reference beam and a sample beam. The ratio between the detected sample intensity and reference intensity was then used to determine the transmission spectrum.

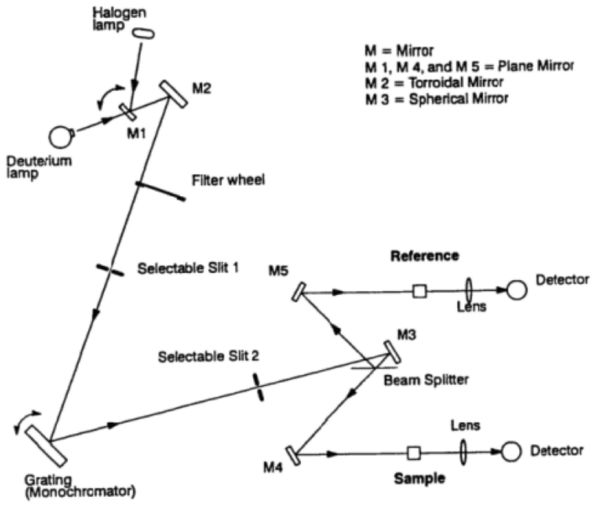


Figure 10: Beam path of the Lambda 40 UV/VIS spectrometer. The incoming light was wavelength-selected by the monochromator and split into a sample beam and a reference beam before detection.

Before each sample measurement, an Autozero or reference measurement was carried out with the appropriate empty sample holder or reference configuration. This reference measurement set the baseline transmission to approximately 100 %. The subsequent measurement with the sample in the beam path yielded the wavelength-dependent transmission given by Eq: 1

The measurements were controlled using the UVWinLab software. The relevant scan parameters included the start and end wavelengths, data interval, scanning speed, slit width, ordinate type, file name, and sample comments. The spectra were exported for further analysis using external software.

3.2 Wavelength calibration and instrumental settings

The wavelength accuracy of the spectrometer was checked using a holmium oxide calibration filter. Holmium oxide has characteristic absorption bands at well-defined wavelengths and was therefore suitable for testing the wavelength scale of the instrument. The measured positions of the absorption minima were compared with reference wavelengths.

Table 1: Reference absorption wavelengths of the holmium oxide filter used for wavelength calibration.

Line	Reference wavelength / nm
1	279.3
2	360.8
3	536.4

The influence of scanning speed was investigated by recording spectra at scan speeds of 240 nm/min, 480 nm/min, 960 nm/min, and 1920 nm/min, while the slit width was kept fixed at 0.5 nm. These spectra were compared in order to determine whether faster scans led to broadening, distortion, or shifts of the absorption minima.

The influence of slit width was investigated by recording spectra with slit widths of 0.5 nm, 1.0 nm, and 2.0 nm, while the scanning speed was kept fixed at 480 nm/min. The resulting spectra were compared in terms of peak depth, peak width, and spectral resolution. Based on these tests, a scan speed of 480 nm/min and a slit width of 0.5 nm were selected for the remaining measurements, since this combination provided a good compromise between measurement time and spectral resolution.

Table 2: Instrumental settings investigated during the holmium oxide calibration measurements.

Measurement series	Varied parameter	Fixed parameter
Scan-speed test	240 nm/min, 480 nm/min, 960 nm/min and 1920 nm/min	Slit width 0.5 nm
Slit-width test	0.5 nm, 1.0 nm and 2.0 nm	Scan speed 480 nm/min
Final calibration scan	Selected settings	480 nm/min, 0.5 nm

3.3 Transmission measurements of cuvettes and optical filters

Transmission spectra of glass and quartz cuvettes were recorded over the wavelength range from approximately 200 nm to 1100 nm. Both cuvette types were measured empty, with air inside the optical path, and filled with distilled water. Before each measurement, a reference measurement was performed using the corresponding empty reference configuration. The spectra were then compared in order to evaluate the ultraviolet, visible, and near-infrared transmission behaviour of glass, quartz, and water.

The comparison between glass and quartz was especially relevant in the ultraviolet region. Ordinary glass absorbs strongly at shorter wavelengths, while quartz remains transparent deeper into the ultraviolet. The water-filled cuvettes were used to identify absorption features originating from water, especially in the near-infrared region.

Transmission spectra of red, green, and blue colour filters were also recorded. The wavelength of maximum transmission was determined for each filter and compared with the spectral region associated with its perceived colour. These measurements were used as a qualitative demonstration of the relation between spectral transmission and colour perception.

3.4 Angle-dependent transmission of the interference filter

The transmission spectrum of an interference filter was measured at several angles of incidence. The filter was placed in the sample beam and rotated to incidence angles of approximately 0°, 15°, 30°, and 45°. For each angle, a new reference or zeroing measurement was performed before the sample scan. This was necessary because rotating the filter changed the optical alignment and the overall transmitted intensity.

The central wavelength of the main transmission band was determined for each angle. The angle dependence was then analysed qualitatively by comparing the measured peak posi-

tions. The expected behaviour was a shift of the transmission maximum toward shorter wavelengths with increasing angle of incidence, because the effective optical path length inside the interference filter decreases at oblique incidence.

Table 3: Incidence angles used for the interference-filter measurements.

Measurement	Angle of incidence
1	0°
2	15°
3	30°
4	45°

3.5 Visible spectral range

The visible spectral range of the eye was estimated by direct observation of the monochromator output. A planar mirror was placed in the measurement beam so that the monochromatic light could be observed directly. For safety reasons, the deuterium lamp was switched off during this part of the experiment, since ultraviolet radiation must not be viewed directly.

The wavelength was varied through the visible region, and the approximate lower and upper limits at which light could still be perceived were recorded. Since this method depended on the observer's eye sensitivity, ambient lighting, and beam intensity, the obtained limits were treated as approximate rather than as sharply defined physical boundaries.

3.6 Iodine vapour spectroscopy

The transmission spectrum of iodine vapour was measured using a glass cuvette containing iodine vapour. The cuvette was placed in the sample beam of the Lambda 40 spectrometer, and a reference measurement was used for baseline correction. The spectrum was recorded over a broad wavelength range, while the analysis focused mainly on the visible region between approximately 500 nm and 650 nm, where the vibronic absorption structure of iodine was observed.

The transmission minima were identified as absorption maxima. These minima corresponded to electronic-vibrational transitions from the electronic ground state of I_2 to vibrational levels of an excited electronic state. The measured wavelengths were converted into photon energies using Eq: 12

For the convergence analysis, the assigned transition energies were plotted as a function of the upper-state vibrational quantum number n'. The decrease in line spacing with increasing n' was then used to determine the convergence energy, dissociation energies, and electronic transition energy according to the equations introduced in Sec. 2.6.

3.7 Generation and measurement of F-centres in KBr

F-centres were generated in a plane-parallel KBr crystal by irradiation with ultraviolet light from a deuterium lamp. The crystal was handled carefully because KBr is hygroscopic. Direct contact with bare hands, breathing on the crystal, or attempts to clean the surface were avoided in order to prevent damage or surface contamination.

After UV irradiation, transmission spectra were recorded at different times in order to follow the time dependence of the F-centre absorption band. In the present analysis, spectra were evaluated directly after irradiation and after approximately 11 min, 34 min, and 45 min. The broad absorption feature in the visible region was assigned to F-centres generated by the irradiation.

Since a separate pre-irradiation reference spectrum was not available for the evaluated KBr data, a slowly varying background transmission was fitted outside the F-centre absorption region and used as an effective reference spectrum. The absorption coefficient was then calculated using

$$\alpha(\lambda) = \frac{1}{d} \ln\left(\frac{T_0(\lambda)}{T(\lambda)}\right), \tag{52}$$

where d is the crystal thickness, $T(\lambda)$ is the measured transmission after irradiation, and $T_0(\lambda)$ is the fitted background transmission. The absorption maximum was used to determine the excitation energy, while the full width at half maximum was used to determine the linewidth. The integrated absorption was used to estimate the concentration of F-centres.

Table 4: Measurement times used for the KBr F-centre spectra after UV irradiation.

Spectrum	Time after irradiation / min
1	0
2	11
3	34
4	45

3.8 Measurement of F-centres in NaCl

F-centres in NaCl were produced by X-ray irradiation. A second non-irradiated NaCl crystal was used as a comparison sample. The irradiated and non-irradiated crystals were measured under the same spectrometer settings so that the additional absorption caused by the colour centres could be isolated.

The NaCl crystals were handled carefully because NaCl is hygroscopic. Direct contact with the crystal surfaces was avoided. The transmission spectrum of the irradiated crystal was compared with the reference spectrum of the non-irradiated crystal. The resulting absorption coefficient was calculated in the same general way as for KBr,

$$\alpha(\lambda) = \frac{1}{d} \ln\left(\frac{T_{\mathrm{ref}}(\lambda)}{T_{\mathrm{irr}}(\lambda)}\right), \tag{53}$$

where $T_{\mathrm{ref}}(\lambda)$ is the transmission of the non-irradiated reference crystal and $T_{\mathrm{irr}}(\lambda)$ is the transmission of the irradiated crystal. The wavelength of maximum absorption, excitation energy, linewidth, integrated absorption, and F-centre concentration were then determined from the absorption band.

Table 5: Crystal parameters used for the F-centre analysis.

Material	Refractive index n	Reflectivity R	Lattice constant a / Å
NaCl	1.544	0.0457	5.63
KBr	1.599	0.0477	6.59

3.9 F-centre concentration analysis

The F-centre concentration was calculated from the integrated absorption band. After the absorption coefficient had been determined as a function of wavelength or wavenumber, the F-centre band was integrated over the selected absorption region. The integral absorption was then related to the F-centre concentration using Eq: 49

The main analysis steps were therefore: conversion of transmission into absorption coefficient, baseline correction where necessary, extraction of the absorption maximum, determination of the full width at half maximum, numerical integration of the absorption coefficient, and calculation of the F-centre concentration.

3.10 Comparison with classical and quantum F-centre models

The experimentally determined F-centre excitation energies were compared with two simple theoretical models. In the classical model, the trapped electron was treated as a harmonic oscillator in the electric field of an effective positive charge distribution. The corresponding excitation energy was calculated using Eq. (38). In the quantum-mechanical model, the trapped electron was treated as a particle in a cubic potential box, and the excitation energy was calculated using Eq. (46).

For both KBr and NaCl, the lattice constants listed in Table 5 were used as input parameters. The calculated excitation energies were then compared with the experimental values obtained from the absorption maxima. In addition, Eq. (47) was used to estimate an effective lattice constant from the measured excitation energy in the quantum box model.

3.11 Uncertainty treatment

The main uncertainty sources were the finite wavelength accuracy of the spectrometer, the wavelength sampling interval, the determination of absorption minima and maxima, baseline subtraction, uncertainty in the crystal thickness, and uncertainty in the material parameters used for the F-centre concentration calculation. For the iodine analysis, an additional uncertainty arose from the assignment of vibronic absorption lines, especially because only a limited number of absorption minima were clearly resolved in the measured spectrum.

For an energy calculated from a measured wavelength, uncertainty propagation gives

$$u(E) = \frac{hc}{\lambda^2} u(\lambda), \tag{54}$$

or, when using energies in electronvolts and wavelengths in nanometres,

$$u(E, /, \mathrm{eV}) = \frac{1239.841984}{(\lambda, /, \mathrm{nm})^2} u(\lambda, /, \mathrm{nm}). \tag{55}$$

For the absorption coefficient calculated using Eq. (5), the uncertainty was estimated from

$$u(\alpha) = \sqrt{\left(\frac{u(T)}{dT}\right)^2 + \left(\frac{\ln T}{d^2} u(d)\right)^2}. \tag{56}$$

For fitted peak positions, absorption maxima, and linewidths, the uncertainty from the fit or graphical extraction was combined with the estimated instrumental wavelength uncertainty. For quantities obtained from numerical integration, the uncertainty was dominated by the baseline choice, the selected integration limits, and the signal-to-noise ratio of the absorption band.

4. Results and Discussion

4.1 Holmium oxide calibration

4.1.1 Measurement procedure

A holmium oxide calibration filter was measured with the Lambda 40 UV/VIS spectrometer in the wavelength range from 200 nm to 1100 nm. Before each measurement, a reference spectrum without the filter was recorded and used for automatic baseline correction.

To investigate the influence of the scanning speed, spectra were recorded using scan speeds of 240, 480, 960 and 1920 nm/min while keeping the slit width fixed at 0.5 nm. Subsequently, the influence of the slit width was studied by measuring spectra with slit widths of 0.5, 1.0 and 2.0 nm at a constant scan speed of 480 nm/min.

The positions of the characteristic holmium oxide absorption minima were determined from the measured transmission spectra and compared with the tabulated reference wavelengths of 279.3, 360.8 and 536.4 nm. The differences between measured and reference values were used to estimate the wavelength accuracy of the spectrometer.

4.1.2 Results

To determine suitable measurement parameters for the remainder of the experiment, the influence of scanning speed and slit width on the recorded spectrum was investigated first.

Influence of scan speed Transmission spectra were recorded using scan speeds of 240, 480, 960 and 1920 nm/min while keeping the slit width fixed at 0.5 nm. The resulting spectra are shown in Fig. 11.

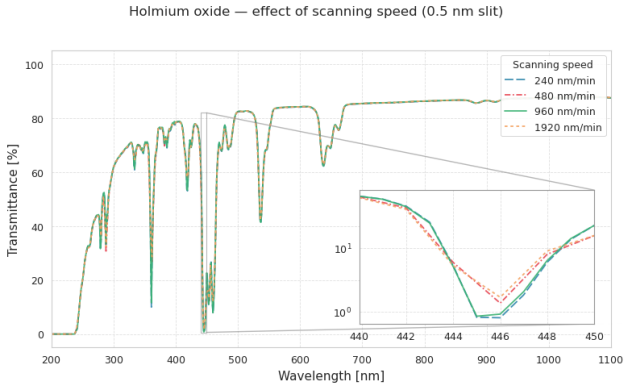


Figure 11: Influence of scan speed on the measured transmission spectrum of the holmium oxide filter. The inset shows a magnified view of the absorption feature near 446 nm.

Only minor differences are observed between the spectra. The measurements recorded at 240 and 480 nm/min are nearly identical, whereas slightly broader features appear at the highest scan speeds. Since a scan speed of 480 nm/min provides essentially the same spectral resolution as 240 nm/min while reducing the measurement time by approximately a factor of two, it was selected for all subsequent measurements.

Influence of slit width The influence of slit width was investigated using slit widths of 0.5, 1.0 and 2.0 nm at a fixed scan speed of 480 nm/min. The corresponding spectra are shown in Fig. 12.

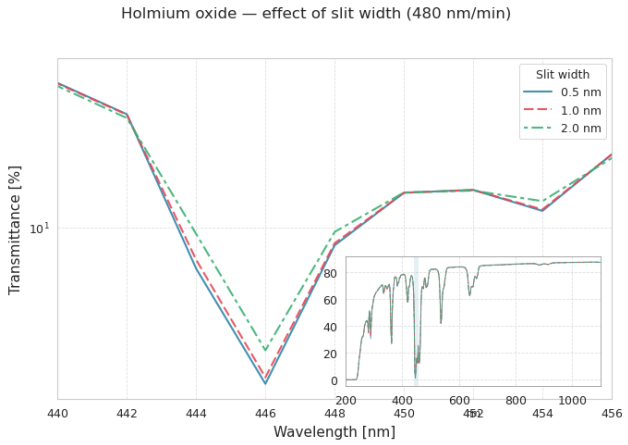


Figure 12: Influence of slit width on the measured transmission spectrum. Smaller slit widths produce narrower and deeper absorption features, corresponding to improved spectral resolution.

Increasing the slit width broadens the absorption features and reduces their depth. The 0.5 nm slit yields the narrowest and most pronounced minima and therefore provides the highest spectral resolution. Consequently, a slit width of 0.5 nm was chosen for the remainder of the experiment.

Wavelength calibration Using the selected settings of 480 nm/min scan speed and 0.5 nm slit width, a final transmission spectrum of the holmium oxide calibration filter was recorded. The identified absorption minima are shown in Fig. 13.

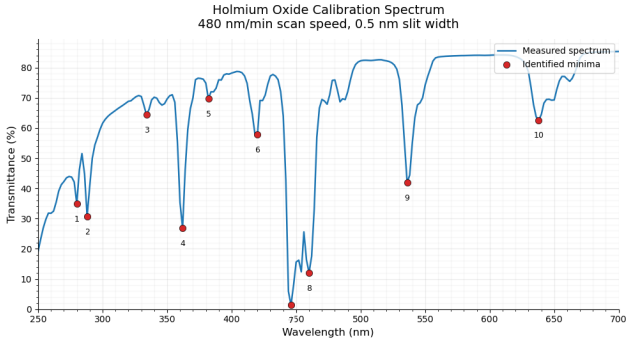


Figure 13: Transmission spectrum of the holmium oxide filter measured using the selected settings of 480 nm/min scan speed and 0.5 nm slit width. The numbered markers indicate the identified absorption minima used for wavelength calibration.

The measured absorption minima were compared with certified reference values provided for the holmium oxide calibration filter. The results are summarized in Table 6.

Table 6: Measured absorption minima of the holmium oxide filter and comparison with certified reference values.

Band	Reference λ / nm	Measured λ / nm	$T_{\min}$ / %	Error / %
1	279.3	280.0	34.95	0.251
2	287.6	288.0	30.73	0.139
3	333.8	334.0	64.37	0.060
4	360.8	362.0	26.90	0.333
5	385.8	382.0	69.66	0.985
6	418.5	420.0	57.87	0.358
7	453.4	446.0	1.35	1.632
8	459.9	460.0	12.09	0.022
9	536.4	536.0	42.00	0.075
10	637.5	638.0	62.60	0.078

The relative deviation of each identified absorption band is shown in Fig. 14. The average relative error of all ten identified bands was found to be approximately 0.39%.

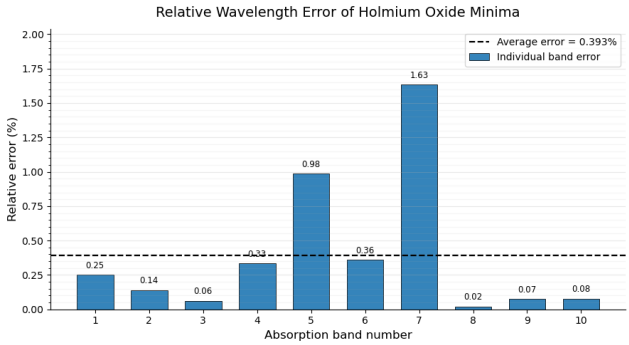


Figure 14: Relative error between measured and certified wavelengths of the identified holmium oxide absorption bands. The dashed line indicates the average relative error of approximately 0.39%.

The measured wavelengths agree well with the certified values of the holmium oxide standard. Most absorption bands show deviations below 0.4%, while the largest deviation of approximately 1.6% occurs for the absorption feature near 453 nm. The average relative error of 0.39% indicates that the wavelength calibration of the spectrometer is accurate within approximately ±1–2 nm over the investigated spectral range.

4.1.3 Discussion

The measured absorption minima agree well with the reference wavelengths of the holmium oxide filter. The deviations are small compared to the spectral range of the instrument and indicate a satisfactory wavelength calibration of the Lambda 40 spectrometer.

To investigate the influence of the scanning speed, spectra were recorded using scan speeds of 240, 480, 960 and 1920 nm/min while keeping the slit width fixed at 0.5 nm. As shown in Fig. 13, all spectra overlap almost perfectly over the entire wavelength range. A magnified view of one of the absorption features reveals only very small differences between the individual measurements. The spectra recorded at 240 and 480 nm/min are nearly indistinguishable, whereas slightly broader minima are observed for the higher scan speeds of 960 and 1920 nm/min. Since the spectral resolution obtained at 480 nm/min is essentially identical to that obtained at 240 nm/min, while requiring only half the measurement time, a scanning speed of 480 nm/min was selected for all subsequent measurements.

The influence of slit width was investigated using slit widths of 0.5, 1.0 and 2.0 nm at a fixed scanning speed of 480 nm/min. The measurements show that increasing the slit width causes the absorption minima to become broader and less pronounced. The 0.5 nm slit produces the deepest and narrowest absorption features, indicating the highest spectral resolution. Although the peak positions remain essentially unchanged, the broader slits reduce the ability to resolve closely spaced spectral features. Therefore, a slit width of 0.5 nm was chosen for the remainder of the experiment.

Overall, the wavelength calibration confirms the correct operation of the spectrometer and the chosen settings of 480 nm/min scan speed and 0.5 nm slit width provide an optimal compromise between measurement time and spectral resolution.

4.2 Cuvettes, water, colour filters and interference filter

4.2.1 Measurement procedure

Transmission spectra of glass and quartz cuvettes were recorded both empty (air-filled) and filled with distilled water in the wavelength range from 200 nm to 1100 nm. The measurements were used to compare the transmission properties of the cuvette materials and to identify absorption bands originating from water.

The transmission spectrum of an interference filter was measured for several angles of incidence (0°, 15°, 30°, and 45°). The position of the transmission maximum was determined for each angle to investigate the angular dependence of the filter.

In addition, transmission spectra of red, green and blue colour filters were recorded. The wavelengths corresponding to the maximum transmission of each filter were determined and compared with the expected spectral ranges associated with the perceived filter colours.

Finally, the visible spectral range was estimated by direct observation of the monochromator output while scanning through the visible wavelength region.

4.2.2 Glass and quartz cuvettes

Figure 15 and 16 shows the transmission spectra of glass and quartz cuvettes filled with air and water.

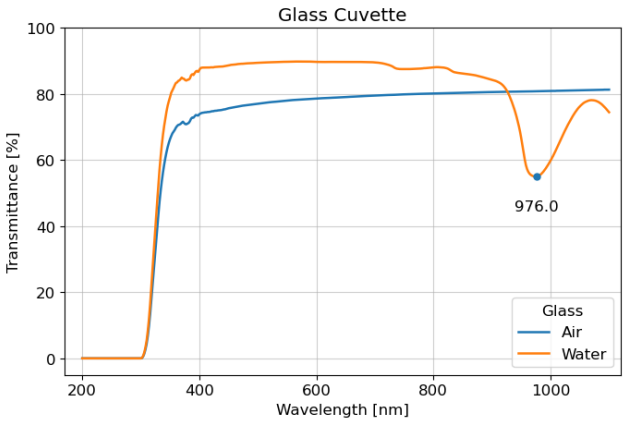


Figure 15: Transmission spectrum of a glass cuvette filled with air and water.

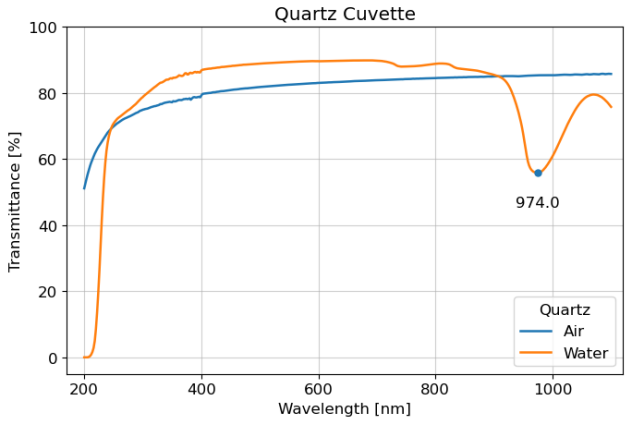


Figure 16: Transmission spectrum of a quartz cuvette filled with air and water.

The main quantitative results are summarized in Table 7.

Table 7: Comparison of transmission properties of glass and quartz cuvettes.

Sample	Mean transmission (400–700 nm) / %	Water absorption minimum / nm	Transmission at minimum / %
Glass (air)	77.55	—	—
Glass (water)	89.34	976	54.98
Quartz (air)	82.25	—	—
Quartz (water)	89.06	974	55.81

Both cuvette types show high transmission in the visible wavelength range. The quartz cuvette exhibits significantly improved transmission in the ultraviolet region compared with ordinary glass, which is expected because quartz remains transparent at shorter wavelengths where glass becomes strongly absorbing.

When the cuvettes are filled with water, an additional absorption feature appears near 975 nm. This absorption band originates from vibrational overtones of water molecules and is observed in both glass and quartz cuvettes. The position of the absorption minimum is consistent with known near-infrared water absorption bands.

4.2.3 Interference filter

The transmission spectra of the interference filter measured at different angles of incidence are shown in Fig. 17.

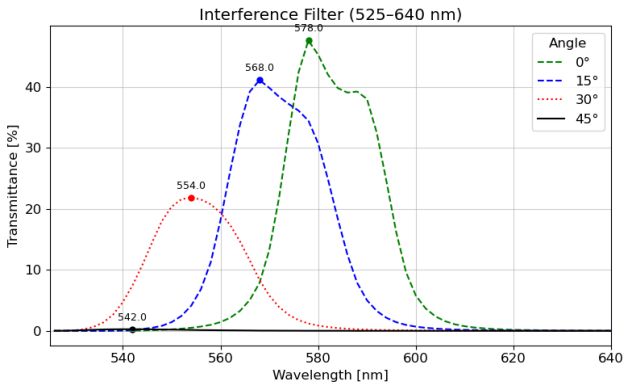


Figure 17: Transmission spectra of the interference filter for different angles of incidence in the visible wavelength range.

The measured peak wavelengths and corresponding maximum transmission values are listed in Table 8.

Table 8: Angle dependence of the interference-filter transmission maximum.

Angle / degree	Peak wavelength / nm	Maximum transmission / %
0	578	47.59
15	568	41.09
30	554	21.86
45	542	0.26

A second transmission band was observed in the near-infrared region for larger incidence angles, as shown in Fig. 18.

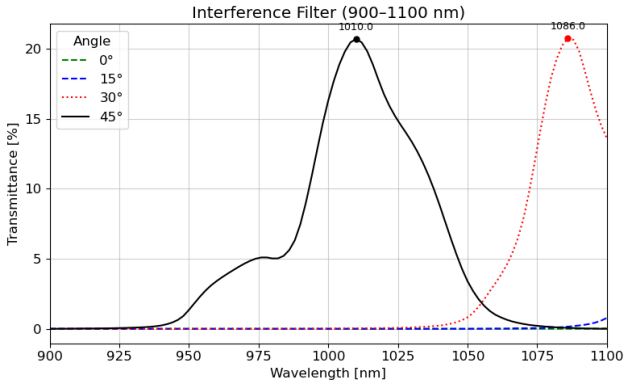


Figure 18: Near-infrared transmission band of the interference filter.

Table 9: Observed near-infrared transmission maxima.

Angle / degree	Peak wavelength / nm	Maximum transmission / %
30	1086	20.74
45	1010	20.70

The transmission maximum shifts systematically toward shorter wavelengths as the angle of incidence increases. The peak wavelength decreases from approximately 578 nm at normal incidence to approximately 542 nm at an incidence angle of 45°. This behaviour is characteristic of interference filters and arises because the effective optical path length inside the dielectric multilayer structure decreases at oblique incidence. Consequently, the constructive-interference condition is fulfilled at shorter wavelengths, producing a blue shift of the transmission maximum.

Furthermore, the maximum transmission decreases significantly with increasing angle, indicating that the filter becomes less efficient away from normal incidence.

4.2.4 Colour filters

The transmission spectra of the red, green and blue colour filters are shown in Fig. 19.

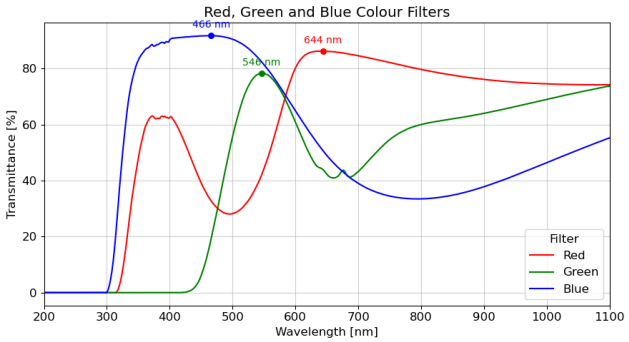


Figure 19: Transmission spectra of the red, green and blue colour filters.

For clarity, the visible wavelength region is shown separately in Figx. 20.

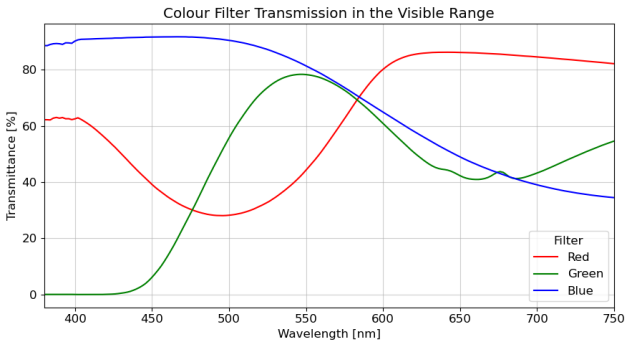


Figure 20: Transmission spectra of the colour filters restricted to the visible wavelength range.

The measured transmission maxima are summarized in Table 10.

Table 10: Transmission maxima of the colour filters.

Filter	Peak wavelength / nm	Maximum transmission / %
Red	644	86.23
Green	546	78.36
Blue	466	91.73

The transmission maxima correspond closely to the expected wavelength ranges associated with red, green and blue light. The blue filter exhibits its maximum transmission near 466 nm, the green filter near 546 nm, and the red filter near 644 nm. These values lie within the spectral regions typically associated with the corresponding perceived colours.

4.2.5 Visible spectral range

Table 11: Estimated visible spectral range from direct observation of the monochromator output.

Quantity	Wavelength / nm
Lower visible limit	380 nm
Upper visible limit	750 nm

The observed visible range is in reasonable agreement with the commonly quoted range of approximately 400 nm to 700 nm. Small deviations are expected because the determination is subjective and depends on the sensitivity of the observer's eye, ambient illumination conditions, and the intensity of the monochromator output. In particular, the eye remains weakly sensitive beyond the conventional limits, making the precise boundaries difficult to define experimentally.

4.3 Iodine vapour spectrum

4.3.1 Measurement procedure

The transmission spectrum of iodine vapour was recorded with the Lambda 40 UV/VIS spectrometer in the wavelength range from approximately 200 nm to 1100 nm. The iodine vapour cell was placed in the sample beam and a reference measurement [2]was used for the baseline correction. The strongest vibronic absorption structure appears in the visible wavelength range between approximately 500 nm and 650 nm. In this region, the transmission curve contains a sequence of closely spaced absorption minima.

Because the absolute assignment of the vibrational quantum number n' is difficult from the measured spectrum alone, reference values for the $n'' = 0 \rightarrow n'$ progression of iodine were used for the line assignment. The measured spectrum was therefore used to verify the absorption region and the presence of the iodine vibronic structure, while the extrapolation was performed using the assigned transition energies.

4.3.2 Measured spectrum and line assignment

The measured transmission spectrum of iodine vapour is shown in Fig. 21. The inset shows the region used for the line assignment. The absorption minima correspond to transitions from the vibrational ground state of the electronic ground state, $n'' = 0$, to vibrational levels n' of the electronically excited state.

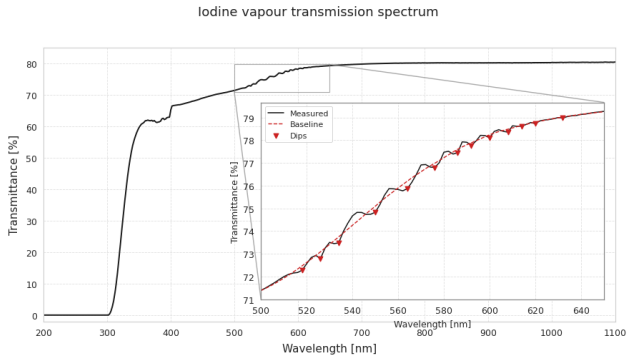


Figure 21: Measured transmission spectrum of iodine vapour. The inset shows the vibronic absorption structure between approximately 500 nm and 650 nm.

The assigned $n'' = 0 \rightarrow n'$ transition energies used for the extrapolation are listed in Table 12[2]. The transition energies were obtained from the wavenumbers using

$$E = \frac{\tilde{\nu}}{8065.54}\,\text{eV},$$

where $\tilde{\nu}$ is given in cm^{-1}.

Table 12: Assigned iodine absorption lines for the $n'' = 0 \rightarrow n'$ progression and corresponding transition energies.

n'	$\tilde{\nu}$ / cm^{-1}	λ / nm	E / eV
14	17301.0	578.00	2.1451
15	17398.9	574.75	2.1572
16	17507.0	571.20	2.1706
17	17602.5	568.10	2.1824
18	17699.1	565.00	2.1944
19	17792.3	562.04	2.2060
20	17883.3	559.18	2.2172
21	17974.0	556.36	2.2285
22	18066.3	553.52	2.2399
23	18146.2	551.08	2.2498
24	18229.5	548.56	2.2602
25	18311.0	546.12	2.2703
26	18395.2	543.62	2.2807
27	18469.3	541.44	2.2899
28	18543.9	539.26	2.2992

4.3.3 Convergence analysis

For the convergence analysis, the transition energies were fitted using the anharmonic oscillator expression for transitions from $n'' = 0$,

$$E(n') = E_{\text{electr}} + E'_S\left(n' + \frac{1}{2}\right) - E'_S y'_e\left(n' + \frac{1}{2}\right)^2.$$

Here E_{electr} is the electronic transition energy, E'_S is the vibrational energy spacing of the excited electronic state, and y'_e is the anharmonicity constant. The value

$$E'_S = 0.0159\,\text{eV}$$

was used for iodine.

The fit and extrapolation to the convergence limit are shown in Fig. 22.

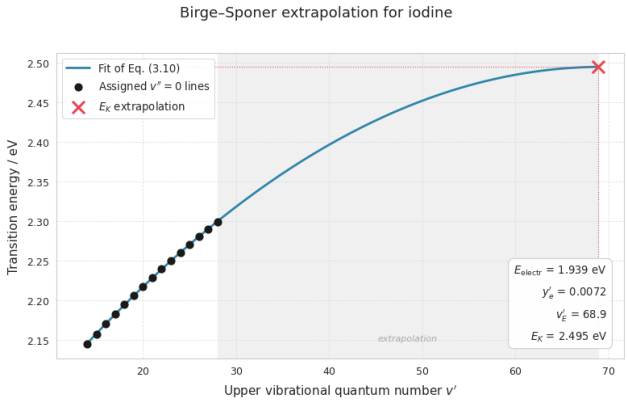


Figure 22: Birge–Sponer extrapolation for iodine. The assigned $n'' = 0 \rightarrow n'$ transition energies were fitted with the anharmonic oscillator expression. The extrapolated maximum gives the convergence energy E_K.

From the fit, the electronic transition energy and anharmonicity constant were obtained as

$$E_{\text{electr}} = (1.9389 \pm 0.0003)\,\text{eV}, \qquad y'_e = 0.00715 \pm 0.00004.$$

The convergence vibrational quantum number follows from

$$n'_E = \frac{1}{2y'_e} - 1,$$

which gives

$$n'_E \approx 68.9.$$

Substituting this value into the fitted energy expression gives the convergence energy

$$E_K = 2.4947\,\text{eV}.$$

Using the iodine atomic excitation energy

$$E_A = 0.94\,\text{eV},$$

the dissociation energies of the ground and excited electronic states are obtained from

$$E''_D = E_K - E_A, \qquad E'_D = E_K - E_{\text{electr}}.$$

The final results are summarized in Table 13.

Table 13: Molecular energies determined from the iodine vapour spectrum.

Quantity	Value
Electronic transition energy E_{electr}	(1.9389 ± 0.0003) eV
Anharmonicity constant y'_e	0.00715 ± 0.00004
Convergence quantum number n'_E	68.9
Convergence energy E_K	2.4947 eV
Ground-state dissociation energy E''_D	1.5547 eV
Excited-state dissociation energy E'_D	0.5558 eV

4.3.4 Discussion

The iodine vapour spectrum shows a broad increase in transmission with superimposed narrow absorption minima in the visible range. These minima arise from vibronic transitions between the electronic ground state and an excited electronic state of the iodine molecule. The line spacing decreases for increasing n', which is expected for an anharmonic molecular potential. In a purely harmonic oscillator, the vibrational levels would be equally spaced, but for a real molecule the level spacing becomes smaller as the dissociation limit is approached.

The extrapolated convergence energy was found to be

$$E_K = 2.4947\,\text{eV}.$$

Together with the known atomic excitation energy of iodine, this gives a ground-state dissociation energy of

$$E_D'' = 1.5547\,\text{eV}.$$

The excited-state dissociation energy is significantly smaller,

$$E_D' = 0.5558\,\text{eV},$$

which is consistent with the fact that the excited electronic state is more weakly bound than the ground state.

The electronic transition energy was determined to be

$$E_{\text{electr}} = 1.9389\,\text{eV}.$$

This corresponds to the vertical energy separation between the minima of the two electronic potentials in the Franck–Condon picture.

An additional limitation of the present measurement is the relatively small number of resolved absorption minima compared with high-resolution reference spectra[2] reported in the literature. In the measured spectrum only a limited portion of the expected vibronic progression could be identified reliably, whereas considerably more absorption lines are usually observed for iodine vapour. During the experiment it was noted by the laboratory supervisor that the iodine cell likely contained a lower iodine vapour concentration than normally used. A reduced iodine concentration decreases the optical density of the sample and leads to weaker absorption features, making the higher-order vibronic transitions difficult to resolve above the experimental noise level. This is the most likely explanation for the comparatively small number of observed minima.

Because of the limited number of clearly resolved absorption lines, the assignment of the vibrational quantum numbers was performed using published reference data rather than solely from the measured spectrum. The transition energies used for the Birge–Sponer analysis were taken from the iodine absorption data reported by King and Arnold (“Absorption and Fluorescence of Molecular Iodine”, Journal of Chemical Physics, 1972). The measured spectrum was therefore used primarily to verify the location and overall structure of the iodine absorption band, while the quantitative determination of the molecular constants relied on the reference-assisted line assignment. Consequently, the dominant uncertainty of the analysis arises from the limited spectral information available in the measured spectrum rather than from the fitting procedure itself.

Nevertheless, the decreasing line spacing and the smooth extrapolation indicate that the assigned lines are consistent with the expected anharmonic behaviour of iodine vapour.

4.4 KBr F-centres

4.4.1 Measurement procedure

A KBr crystal was irradiated with UV light in order to generate F-centres. After irradiation, transmission spectra were recorded at different times: directly after irradiation, after 11 min, after 34 min, and after 45 min. The spectra were measured in the wavelength range from approximately 200 nm to 1100 nm. Since no separate transmission spectrum before irradiation was available, the slowly varying background transmission was removed by fitting a smooth baseline outside the F-centre absorption region. The remaining absorption band was then used to determine the excitation energy, linewidth and F-centre concentration.

4.4.2 Results

The measured transmission spectra are shown in Fig. 23. A broad absorption feature is visible in the region around 600 nm to 700 nm, which is attributed to the F-centres generated by UV irradiation.

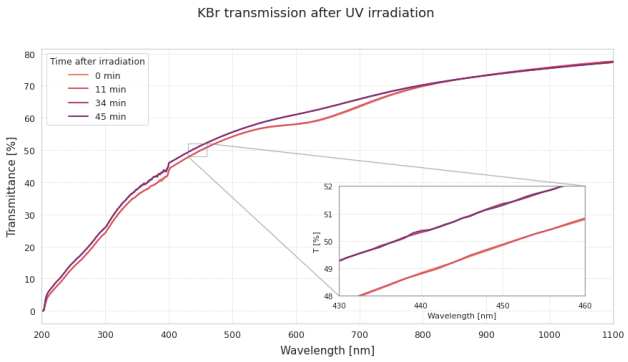


Figure 23: Transmission spectra of the KBr crystal after UV irradiation at different times.

The F-centre absorption region is shown separately in Fig. 24.

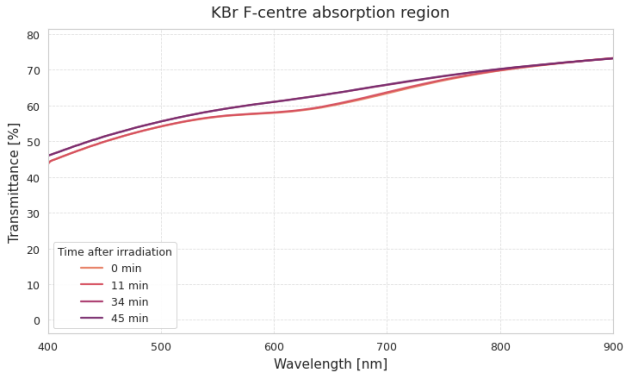


Figure 24: Zoomed view of the KBr F-centre absorption region.

The absorption coefficient was calculated from the baseline-corrected transmission using the Lambert–Beer relation

$$\alpha(\tilde{\nu}) = \frac{1}{d} \ln\left(\frac{T_0(\tilde{\nu})}{T(\tilde{\nu})}\right),$$

where d is the thickness of the KBr crystal, $T(\tilde{\nu})$ is the measured transmission and $T_0(\tilde{\nu})$ is the fitted reference baseline.

The resulting absorption coefficient as a function of wavenumber is shown in Fig. 25.

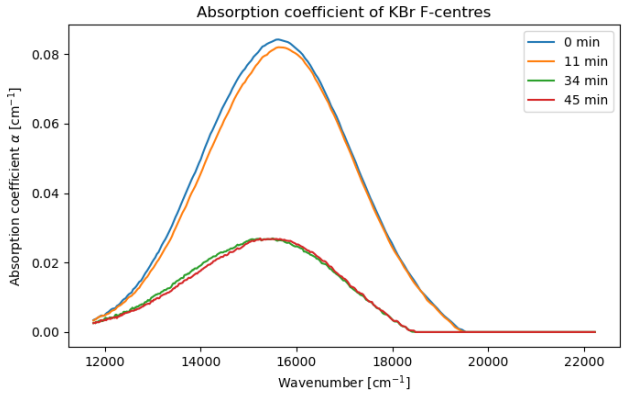


Figure 25: Absorption coefficient of the KBr F-centres as a function of wavenumber. The integral over this curve was used to determine the F-centre concentration.

The F-centre concentration was calculated from the integral absorption using

$$\int \alpha \, d\tilde{\nu} = \frac{e^2 N}{4c^2 \varepsilon_0 n m_e} \left(\frac{n^2+2}{3}\right)^2.$$

Solving for the concentration gives

$$N = \frac{4c^2 \varepsilon_0 n m_e}{e^2} \left(\frac{3}{n^2+2}\right)^2 \int \alpha \, d\tilde{\nu}.$$

For KBr, the refractive index was taken as

$$n = 1.599.$$

The evaluated peak positions, excitation energies, linewidths and concentrations are summarized in Table 14.

Table 14: F-centre absorption parameters and concentration in KBr after UV irradiation.

Time / min	λ_{max} / nm	E_{exc} / eV	FWHM / nm	$\int \alpha d\tilde{\nu}$ / cm^{-2}	N / cm^{-3}
0	640	1.937	153.25	320.62	2.51×10^{14}
11	640	1.937	148.56	306.76	2.40×10^{14}
34	645	1.922	161.00	100.90	7.90×10^{13}
45	656	1.890	154.60	98.92	7.74×10^{13}

The time dependence of the F-centre concentration is shown in Fig. 26.

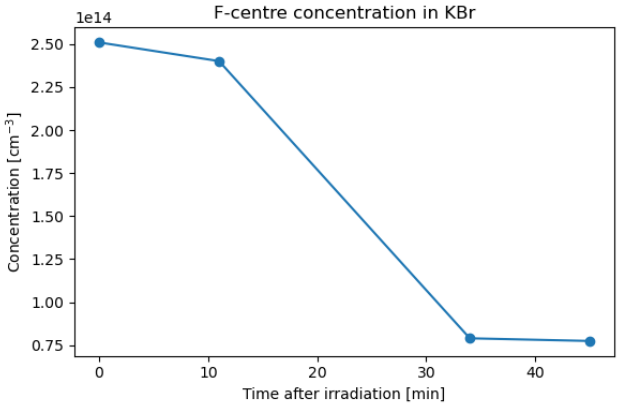


Figure 26: F-centre concentration in KBr as a function of time after UV irradiation.

4.4.3 Discussion

The measured spectra show a broad absorption band in the visible range, which is characteristic of F-centres in KBr. The maximum of the absorption band occurs initially at approximately 640 nm. This corresponds to an excitation energy

$$E = \frac{hc}{\lambda} \approx 1.94\,\text{eV}.$$

This value is physically reasonable for the optical excitation of F-centres in KBr.

The FWHM of the absorption band is approximately 150 nm. The large linewidth is expected because the electronic transition of an F-centre is strongly coupled to lattice vibrations. Due to the Franck–Condon principle, the transition does not occur between two sharp electronic levels only, but involves many vibrational configurations of the lattice. Therefore, a broad absorption band is observed instead of a narrow spectral line.

The concentration decreases from

$$2.51 \times 10^{14}\,\text{cm}^{-3}$$

directly after irradiation to

$$7.74 \times 10^{13}\,\text{cm}^{-3}$$

after 45 min. This decrease is physically plausible because F-centres are not completely stable after irradiation. In the

absence of continued UV exposure, some centres recombine or relax, reducing the number of optically active defects.

However, the results also show some limitations. The strongest decrease occurs between 11 min and 34 min, while the concentrations at 34 min and 45 min are nearly identical. This suggests that the late-time absorption signal is weak and strongly affected by baseline subtraction. The apparent shift of the peak from 640 nm to 656 nm at later times is therefore likely not a real physical shift of the F-centre transition, but rather a consequence of the weaker absorption band and the uncertainty in determining the maximum.

The absolute concentration values are lower than those reported in some reference reports. This may be caused by differences in irradiation time, UV intensity, crystal thickness, baseline correction, or by the absence of a true pre-irradiation reference spectrum. Nevertheless, the qualitative behaviour is correct: the F-centre absorption band is observed, the excitation energy is close to the expected value, and the concentration decreases with time after irradiation.

4.5 NaCl F-centres

4.5.1 Measurement procedure

The NaCl F-centre analysis was performed using transmission spectra of a non-irradiated NaCl crystal and an X-ray irradiated NaCl crystal. The non-irradiated crystal was used as the reference spectrum, while the irradiated crystal contained the additional absorption caused by the generated colour centres. This allowed the absorption coefficient of the radiation-induced defect band to be calculated directly from Eq. (53), without requiring a fitted background reference as in the KBr analysis.

The crystal thickness was taken as (d=4 mm=0.4 cm), and the refractive index (n=1.544) was used for the concentration calculation, as listed in Table 5 [1]. After the absorption coefficient had been calculated, the main F-centre absorption band was isolated from the broader background absorption. The peak wavelength was used to determine the excitation energy using Eq. (32). The full width at half maximum was determined from the fitted main absorption band using Eq. (33), and the F-centre concentration was obtained from the integral absorption using Eq. (49).

4.5.2 Absorption band, excitation energy, line width and concentration

The measured transmission spectra of the non-irradiated and X-ray irradiated NaCl crystals are shown in Fig. 27. The irradiated crystal showed a substantially lower transmission than the non-irradiated reference crystal, especially in the visible spectral range. This additional absorption was attributed to colour centres produced by X-ray irradiation.

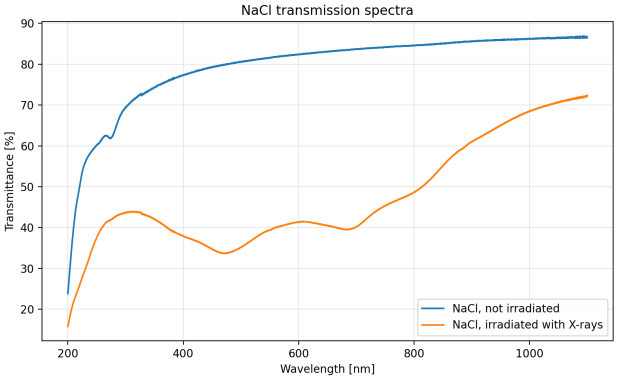


Figure 27: Transmission spectra of the non-irradiated and X-ray irradiated NaCl crystals. The reduction in transmission after irradiation indicates additional absorption caused by radiation-induced colour centres.

The corresponding absorption coefficient, calculated using Eq. (53), is shown over the full measured wavelength range in Fig. 28. The spectrum contains a strong absorption band in the visible region and a broader secondary absorption structure at longer wavelengths.

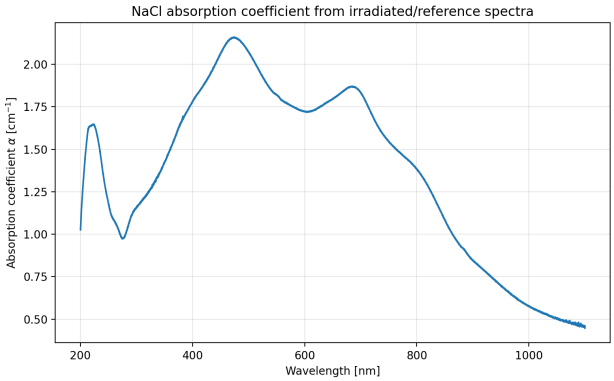


Figure 28: Absorption coefficient of the X-ray irradiated NaCl crystal relative to the non-irradiated reference crystal.

For the quantitative F-centre analysis, the visible absorption region was examined in more detail, as shown in Fig. 29. The main F-centre absorption band was located near 474 nm. A broader secondary absorption contribution was also visible at longer wavelengths and was not treated as the main F-centre band.

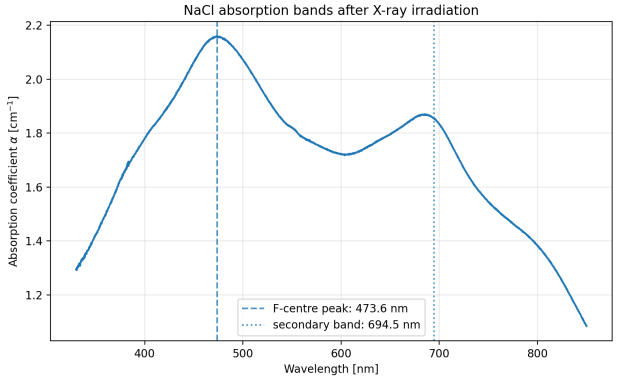


Figure 29: Zoomed view of the NaCl absorption spectrum in the visible and near-infrared region. The main F-centre absorption band appears near 474 nm, while a broader secondary absorption structure occurs at longer wavelengths.

To separate the main F-centre band from the slowly varying background and the secondary broad absorption feature, the absorption spectrum was fitted with a smooth background and two broad band contributions. The resulting fit is shown in Fig. 30. The fitted maximum of the main F-centre band was found at

$$\lambda_m = 473.65\,\text{nm}.$$

Using Eq. (32), this corresponds to an excitation energy of

$$E_m = 2.6176\,\text{eV}.$$

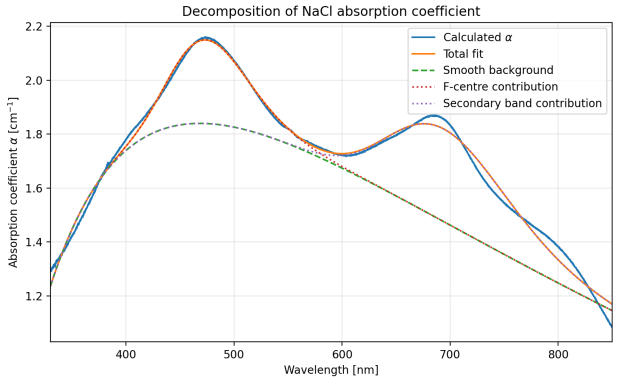


Figure 30: Fit of the NaCl absorption spectrum. The main F-centre band was separated from the slowly varying background and the broader secondary absorption contribution.

The isolated F-centre band used for the integral absorption is shown in Fig. 31. The integration of this band gave

$$\int \alpha, \mathrm{d}\tilde{\nu} = 1230.70\,\text{cm}^{-2}.$$

Using Eq. (49), the corresponding F-centre concentration was calculated as

$$N = 1.01 \times 10^{15}, 1/\text{cm}^3.$$

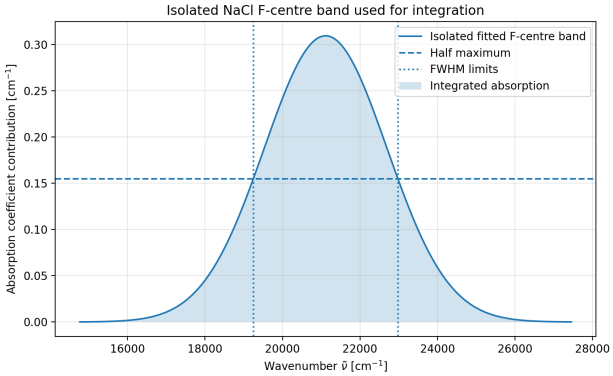


Figure 31: Isolated NaCl F-centre absorption band used for numerical integration. The integral absorption was used to calculate the F-centre concentration.

The final absorption parameters obtained for the X-ray irradiated NaCl crystal are summarized in Table 15.

Table 15: NaCl F-centre properties determined from the fitted absorption band.

Quantity	Value
Peak wavelength λ_m	473.65 nm
Excitation energy E_m	2.6176 eV
FWHM	84.43 nm
FWHM ΔE_H	0.4629 eV
Integral absorption $\int \alpha\,\mathrm{d}\tilde{\nu}$	1230.70 cm^{-2}
F-centre concentration N	1.01×10^{15} $1/\text{cm}^3$

4.5.3 Discussion

The X-ray irradiated NaCl crystal showed a clear F-centre absorption band with a maximum at approximately 474 nm. This wavelength lies in the blue region of the visible spectrum. Since the irradiated crystal absorbed strongly in this part of the spectrum, the transmitted light would be shifted toward the complementary colour region. This explains the characteristic colouring of irradiated alkali halide crystals containing F-centres.

The excitation energy of the NaCl F-centre was found to be 2.6176 eV. This value is larger than the excitation energy found for KBr, which was approximately 1.94 eV. The difference is physically reasonable because NaCl has a smaller lattice constant than KBr, as shown in Table 5. A smaller vacancy region confines the trapped electron more strongly, leading to a larger energy separation between the ground and excited defect states. This trend is consistent with the qualitative prediction of the quantum-mechanical box model discussed in Sec. 2.10.

The fitted linewidth of the NaCl F-centre band was ($\Delta E_H = 0.4629$ eV). The large linewidth confirmed that the transition was not a sharp atomic-like transition, but a broad solid-state absorption band. This broadening was attributed to electron–phonon coupling and the Franck–Condon

mechanism described in Sec. 2.8. During optical excitation, the electronic state of the trapped electron changes faster than the surrounding ions can rearrange. As a result, many vibrational configurations contribute to the absorption process, producing a broad band rather than a narrow line.

The F-centre concentration was determined to be $N = 1.01 \times 10^{15}, 1/\mathrm{cm}^3$. This value is larger than the concentrations obtained for the UV-irradiated KBr crystal. This difference may be caused by the different irradiation method, irradiation dose, crystal material, and the quality of the reference measurement. In contrast to the KBr analysis, the NaCl analysis used a directly measured non-irradiated reference crystal. Therefore, the absolute absorption coefficient and concentration were less dependent on a fitted baseline correction.

The main uncertainty in the NaCl analysis came from separating the main F-centre band from the broader secondary absorption structure at longer wavelengths. The peak position and excitation energy were relatively robust, because the maximum of the main band was clearly located near 474 nm. The linewidth and concentration were more sensitive to the fitted background, the assumed band shape, and the selected integration range. Therefore, the excitation energy should be regarded as the most reliable result, while the linewidth and concentration carry a larger systematic uncertainty.

4.6 Comparison with classical and quantum models

The experimental excitation energies of the F-centres in KBr and NaCl were compared with the classical oscillator model and the simple quantum-mechanical box model. The material constants used for the calculations are listed in Table 16. The classical excitation energies were calculated using Eq. (38), while the quantum-mechanical excitation energies were calculated using Eq. (46). In addition, Eq. (47) was used to estimate the effective lattice constant that would be required for the quantum box model to reproduce the measured excitation energy.

For KBr, the strongest and most reliable absorption band was the spectrum recorded directly after irradiation. The corresponding excitation energy from Table 14 was therefore used for the model comparison. The later KBr spectra were not used for this comparison because the absorption band became weaker with time and the peak position was more strongly affected by the baseline correction. For NaCl, the excitation energy from Table 15 was used.

Table 16: Material constants used for the F-centre model calculations [1].

Material	n	R	a / Å
NaCl	1.544	0.0457	5.63
KCl	1.490	0.0387	6.29
KBr	1.599	0.0477	6.59

Table 17: Comparison between experimental and calculated F-centre excitation energies.

Material	E_m^{exp} / eV	E_{cl} / eV	E_{qm} / eV	$a_{\mathrm{exp,qm}}$ / Å
KBr	1.937	2.131	2.598	7.63
NaCl	2.618	2.699	3.559	6.56

The comparison shows that both models reproduce the correct qualitative trend. The excitation energy of the NaCl F-centre is larger than that of the KBr F-centre. This is expected because NaCl has a smaller lattice constant than KBr, so the trapped electron is more strongly confined. Stronger confinement leads to a larger separation between the ground and excited defect states.

Quantitatively, the classical oscillator model agreed better with the experimental values than the simple quantum box model. For KBr, the classical model gave $E_{\mathrm{cl}} = 2.131\,\mathrm{eV}$, compared with the experimental value $E_m^{\mathrm{exp}} = 1.937\,\mathrm{eV}$. For NaCl, the classical model gave $E_{\mathrm{cl}} = 2.699\,\mathrm{eV}$, compared with the experimental value $E_m^{\mathrm{exp}} = 2.618\,\mathrm{eV}$. The agreement was especially good for NaCl.

The quantum box model overestimated the excitation energies for both materials. It predicted $E_{\mathrm{qm}} = 2.598\,\mathrm{eV}$ for KBr and $E_{\mathrm{qm}} = 3.559\,\mathrm{eV}$ for NaCl. These values are significantly larger than the measured excitation energies. The same conclusion is visible from the effective lattice constants calculated from Eq. (47). The quantum model would require effective box lengths of 7.63 Å for KBr and 6.56 Å for NaCl, both of which are larger than the actual lattice constants listed in Table 16.

This discrepancy is physically reasonable. The simple quantum model assumed an infinitely deep cubic potential box with the side length equal to the lattice constant. In a real F-centre, the electron is not confined by infinitely hard walls and its wave function can extend into the neighbouring lattice region. Therefore, the effective confinement length is larger than the lattice constant, and the real transition energy is lower than the value predicted by the idealized box model.

The classical model also remained simplified because it replaced the discrete ionic environment by a homogeneous positive charge distribution. Nevertheless, it gave a better numerical estimate for the measured transition energies. Overall, the comparison showed that the models captured the main size trend of F-centres in alkali halides, but that a realistic quantitative description would require a more detailed defect potential including lattice polarization and electron–phonon coupling.

5. Conclusion

In this experiment, UV/VIS optical spectroscopy was used to investigate molecular iodine vapour and colour centres in alkali halide crystals. The Lambda 40 spectrometer was first checked using a holmium oxide calibration filter. The selected settings of 480 nm/min scan speed and 0.5 nm slit width provided a suitable compromise between measurement time and spectral resolution. The measured holmium oxide absorption minima agreed well with the reference wavelengths, with an average relative error of approximately 0.39%. This confirmed that the wavelength calibration of the spectrometer was sufficiently accurate for the later spectral analysis.

The transmission measurements of cuvettes and optical filters showed the expected optical behaviour. Quartz transmitted more strongly in the ultraviolet region than ordinary glass, while water-filled cuvettes showed a clear near-infrared absorption feature near 975 nm. The colour filters transmitted light mainly in their corresponding visible spectral regions, with maxima at 466 nm, 546 nm, and 644 nm for the blue, green, and red filters, respectively. The interference filter showed a systematic blue shift of its transmission maximum from 578 nm at normal incidence to 542 nm at an incidence angle of 45°, consistent with the reduction of the effective optical path length at oblique incidence.

The iodine vapour spectrum showed a structured absorption band in the visible region. The measured spectrum confirmed the presence of vibronic absorption features, although only a limited number of absorption minima could be resolved clearly. Using a reference-assisted line assignment and an anharmonic oscillator fit, the electronic transition energy was determined as ($E_{\text{electr}} = (1.9389 \pm 0.0003)$ eV). The convergence energy was found to be ($E_K = 2.4947$ eV), giving a ground-state dissociation energy of ($E_D'' = 1.5547$ eV) and an excited-state dissociation energy of ($E_D' = 0.5558$ eV). The smaller excited-state dissociation energy was consistent with the weaker binding of the electronically excited iodine molecule.

F-centres were successfully observed in both KBr and NaCl. For KBr, UV irradiation produced a broad absorption band near 640 nm, corresponding to an excitation energy of approximately 1.94 eV. The F-centre concentration decreased from 2.51×10^{14} $1/\text{cm}^3$ directly after irradiation to ($7.74 \times 10^{13}, 1/\text{cm}^3$) after 45 min. This decrease indicated a reduction in the number of optically active colour centres after irradiation. However, since no non-irradiated KBr reference spectrum was available, the absolute concentration values depended strongly on the fitted baseline and should therefore be interpreted as estimates.

For the X-ray irradiated NaCl crystal, the F-centre absorption maximum was found at 473.65 nm, corresponding to an excitation energy of 2.6176 eV. The fitted linewidth was ($\Delta E_H = 0.4629$ eV), and the F-centre concentration obtained from the integrated absorption was ($1.01 \times 10^{15}, 1/\text{cm}^3$). The larger excitation energy in NaCl compared with KBr was physically reasonable because the smaller lattice constant of NaCl leads to stronger confinement of the trapped electron.

The comparison with theoretical models showed that both the classical oscillator model and the quantum-mechanical box model reproduced the correct qualitative trend: the F-centre excitation energy increased as the lattice constant decreased. Quantitatively, the classical model agreed better with the experimental values. For KBr, it predicted 2.131 eV compared with the measured 1.937 eV, while for NaCl it predicted 2.699 eV compared with the measured 2.618 eV. The simple quantum box model overestimated both excitation energies because it assumed infinitely hard confinement with a box length equal to the lattice constant. In real F-centres, the electron wave function extends into the surrounding lattice, so the effective confinement length is larger and the transition energy is lower.

Overall, the experiment demonstrated how optical spectroscopy can connect measured transmission spectra with microscopic physical processes. The iodine spectrum illustrated electronic-vibrational coupling in a molecule, while the KBr and NaCl spectra showed how localized defect states in ionic crystals produce broad colour-centre absorption bands. The main limitations were the weakly resolved iodine spectrum, the missing non-irradiated KBr reference, and the model-dependent separation of the NaCl F-centre band from the broader background absorption. Despite these limitations, the obtained results were physically consistent and supported the Franck–Condon interpretation of both molecular vibronic structure and broad F-centre absorption bands.

A. Appendix

A.1 Natural constants

Table 18: Natural constants used in the analysis.

Quantity	Symbol	Value
Speed of light	c	$2.9979 \cdot 10^{8}$ m/s
Electron mass	m_e	$9.1094 \cdot 10^{-31}$ kg
Elementary charge	e	$1.6022 \cdot 10^{-19}$ C
Planck constant	h	$6.6261 \cdot 10^{-34}$ J s
Reduced Planck constant	$\hbar$	$1.0546 \cdot 10^{-34}$ J s
Vacuum permittivity	ε_0	$8.8542 \cdot 10^{-12}$ F/m
Energy conversion	hc	1239.841 984 eV nm
Wavenumber conversion	1 eV	8065.54/cm

A.2 Material parameters for alkali halides

Table 19: Material parameters used for the F-centre analysis and model comparison.

Material	Refractive index n	Reflectivity R	Lattice constant a / Å
NaCl	1.544	0.0457	5.63
KCl	1.490	0.0387	6.29
KBr	1.599	0.0477	6.59

A.3 Iodine line assignment used for the Birge–Sponer analysis

Table 20: Assigned iodine absorption lines for the $n'' = 0 \rightarrow n'$ progression. These values were used for the Birge–Sponer extrapolation in Sec. 4.3.

n'	$\tilde{\nu}$ / 1/cm	λ / nm	E / eV
14	17301.0	578.00	2.1451
15	17398.9	574.75	2.1572
16	17507.0	571.20	2.1706
17	17602.5	568.10	2.1824
18	17699.1	565.00	2.1944
19	17792.3	562.04	2.2060
20	17883.3	559.18	2.2172
21	17974.0	556.36	2.2285
22	18066.3	553.52	2.2399
23	18146.2	551.08	2.2498
24	18229.5	548.56	2.2602
25	18311.0	546.12	2.2703
26	18395.2	543.62	2.2807
27	18469.3	541.44	2.2899
28	18543.9	539.26	2.2992

A.4 Summary of iodine molecular parameters

Table 21: Molecular parameters obtained from the iodine vapour analysis.

Quantity	Value
Electronic transition energy E_{electr}	(1.9389 ± 0.0003) eV
Anharmonicity constant y'_e	0.00715 ± 0.00004
Convergence quantum number n'_E	68.9
Convergence energy E_K	2.4947 eV
Ground-state dissociation energy E''_D	1.5547 eV
Excited-state dissociation energy E'_D	0.5558 eV

A.5 KBr F-centre analysis summary

Table 22: Summary of the KBr F-centre parameters obtained after UV irradiation.

Time / min	$\lambda_{\max}$ / nm	E_{exc} / eV	FWHM / nm	$\int \alpha \, \mathrm{d}\tilde{\nu}$ / cm^{-2}	N / 1/cm^3
0	640	1.937	153.25	320.62	2.51×10^{14}
11	640	1.937	148.56	306.76	2.40×10^{14}
34	645	1.922	161.00	100.90	7.90×10^{13}
45	656	1.890	154.60	98.92	7.74×10^{13}

Since no separate non-irradiated KBr reference spectrum was available, the slowly varying background transmission was estimated by fitting a smooth baseline outside the main F-centre absorption band. Therefore, the KBr concentration values should be treated as baseline-dependent estimates. The trend in concentration with time is more reliable than the absolute concentration values.

A.6 NaCl F-centre analysis summary

Table 23: Summary of the NaCl F-centre parameters obtained from the X-ray irradiated crystal.

Quantity	Value
Peak wavelength λ_m	473.65 nm
Excitation energy E_m	2.6176 eV
FWHM	84.43 nm
FWHM ΔE_H	0.4629 eV
Integral absorption $\int \alpha \, \mathrm{d}\tilde{\nu}$	1230.70 cm^{-2}
F-centre concentration N	$1.01 \cdot 10^{15}$/cm^3

For the NaCl analysis, the non-irradiated crystal spectrum was used directly as the reference spectrum. This made the absorption coefficient and concentration calculation more reliable than for KBr, where the reference transmission had to be estimated from a fitted baseline.

A.7 Classical and quantum model comparison

Table 24: Comparison between experimental and calculated F-centre excitation energies.

Material	E_m^{exp} / eV	E_{cl} / eV	E_{qm} / eV	$a_{\text{exp,qm}}$ / Å
KBr	1.937	2.131	2.598	7.63
NaCl	2.618	2.699	3.559	6.56

The experimental and calculated values showed the same qualitative trend: the NaCl F-centre had a larger excitation energy than the KBr F-centre. Quantitatively, the classical oscillator model agreed better with the experimental values than the simple

quantum-mechanical box model. The box model overestimated the transition energies because it assumed infinitely hard confinement with a box length equal to the lattice constant.

A.8 Use of AI assistance

ChatGPT was used as an auxiliary tool during the preparation of this report. It was used to support the structure of the report, improve wording and clarity, identify inconsistencies in the draft, and provide guidance for data analysis and graph plotting. All experimental data, numerical calculations, plots, final interpretations, and conclusions were checked and verified by the authors. ChatGPT was not used as a primary scientific source for the physical theory or reference values.

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